

CHAPTER 1

PREAMBLE

1.1 Introduction

Modern advanced cameras that can monitor under all circumstances with dependable effectiveness especially in dimly lit areas are in greater demand as security threats develop. Due to their inability to provide precise and ongoing monitoring in the dark, traditional surveillance systems may be vulnerable to security breaches. The "Night Vision Motion Detection" project was created to improve security by combining automated alarm systems, real-time motion detection, and night vision technologies in order to address these limitations. The system's goal is to guarantee a high level of security in a variety of settings by offering prompt alerts and continuous observation, independent of lighting conditions.

A night vision camera, which can record and broadcast live video feeds even in total darkness, is the main component of the project. OpenCV, a potent and extensively used open-source computer vision toolkit, is utilized to process this real-time video stream. OpenCV uses sophisticated algorithms to accurately identify human motion while removing background movements that aren't important, such wind or passing animals. By reducing the possibility of false alarms, this intelligent filtering makes sure that the system only sounds an alert in response to real dangers.

The device plays a pre-recorded audio file as soon as it detects human presence. This ensures that people in the vicinity or security staff are instantly informed to possible incursions or security breaches. In order to improve security measures' efficacy and shorten response times in the event of unauthorized access or questionable activity, this automated alarm system is essential. The "Night Vision Motion Detection" project offers a dependable and all-inclusive monitoring solution for a variety of settings, including residential, commercial, and industrial environments where security is of the utmost importance. This is achieved by combining night vision capabilities with advanced motion detection algorithms.

This project uses modern technology to improve security in low light, provide continuous monitoring, lower alarm sounds, and provide prompt reactions to possible attacks. It is a big step forward in these areas.

Night Vision Motion Detection

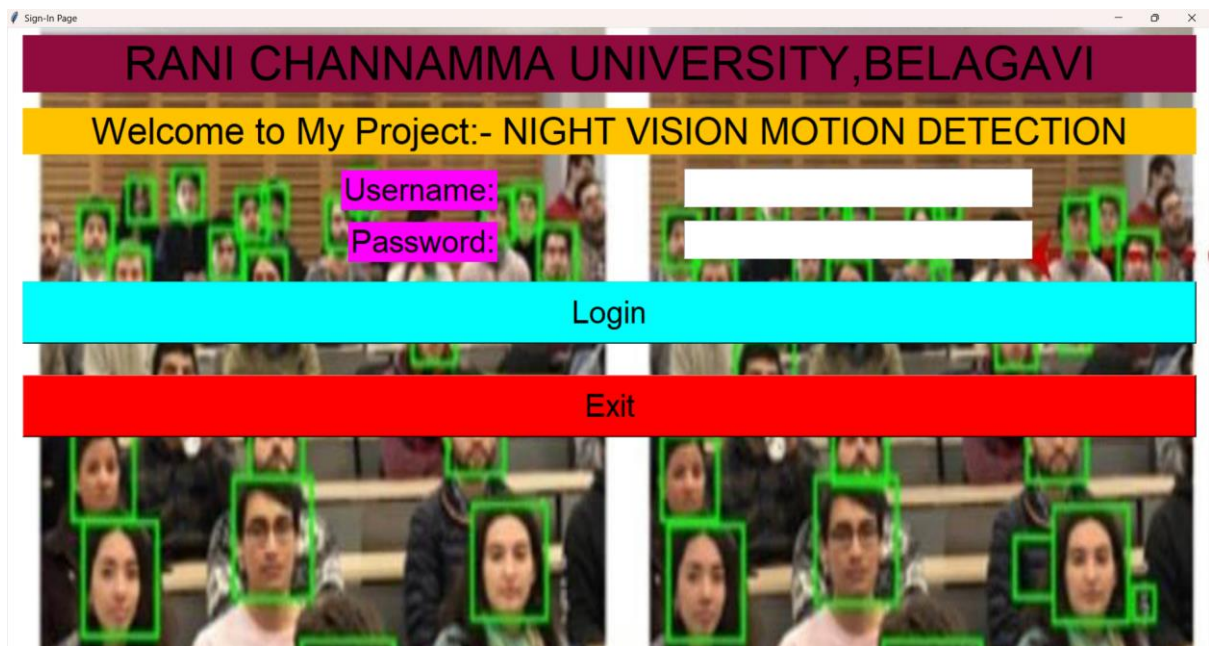


Figure 1:- Sign in Page

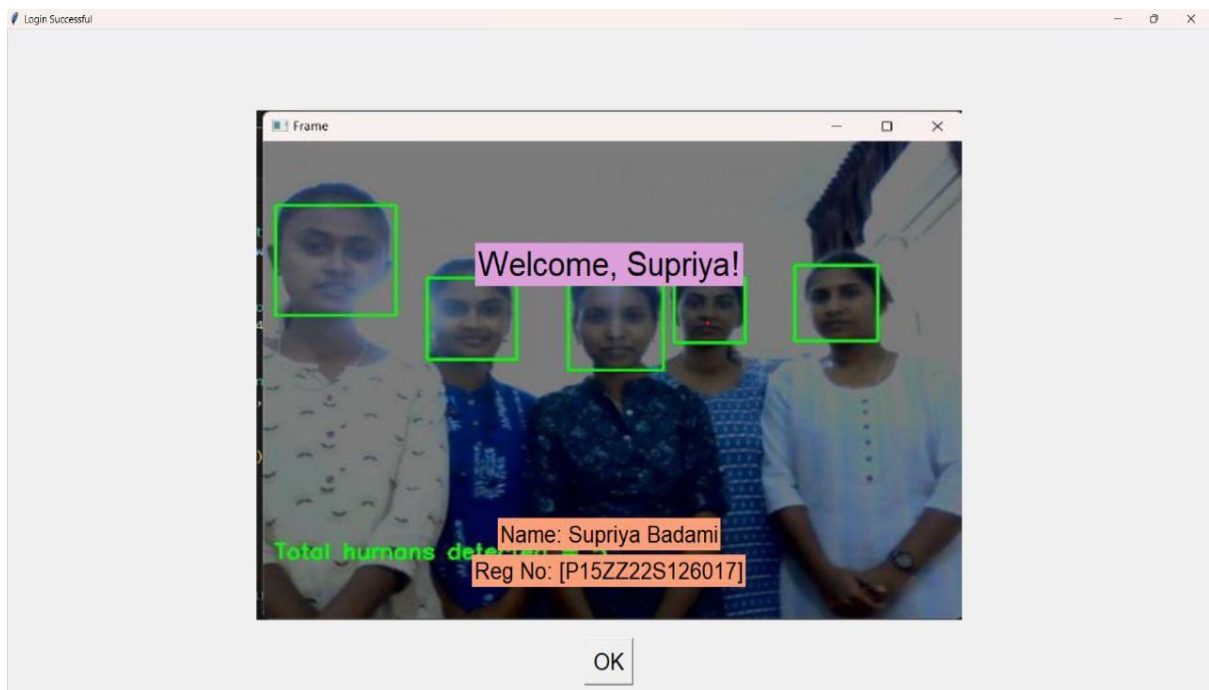


Figure 1.2:- Login with Frame Output

1.2 Objectives

- The "Night Vision Motion Detection" project's primary goal is to improve security by offering trustworthy and ongoing surveillance, particularly in low light or total darkness.
- Real-time motion detection and night vision technology are combined by the system to detect human presence while reducing false alarms from background movements that aren't relevant.
- The system efficiently detects human movements and sets off an automated alarm using a night vision camera and the OpenCV library, ensuring prompt alerts and quick reactions to possible threats.
- The mainl of this project is to provide a reliable and effective monitoring system for a range of environments, guaranteeing security and safety all the time.

1.3 Motivation

- The "Night Vision Motion Detection" project was inspired by the growing demand for reliable security solutions in settings where conventional surveillance systems frequently fall short, especially at night or in low light.
- Because many of the devices in place are unable to distinguish between background activity that is irrelevant and human movement, they frequently experience poor visibility and high false alarm rates.
- By combining cutting-edge motion detection algorithms with night vision technology, this initiative seeks to address these issues and provide dependable threat identification and monitoring regardless of illumination.
- The system gives customers in residential, commercial, and industrial settings better overall security and peace of mind by lowering false alarms and increasing detection accuracy.

CHAPTER 2

LITERATURE SURVEY

2.1 Existing System

Existing monitoring systems are severely limited, especially at night or in poor light. Conventional security cameras work well in well-lit areas, but they frequently can't detect motion or produce sharp images at night. These systems usually use low-lux or infrared (IR) cameras, which have limited clarity and range. As a result, they are not able to distinguish between human movement and other background activities like animals, shadows, or motion caused by the weather, which leads to poor visibility and a high rate of false alarms.

In addition, many of the systems currently in use are incapable of intelligently identifying and classifying motions that aren't relevant, which frequently leads to false alerts that could cause unwarranted worry and reduce the effectiveness of security measures. Basic motion detection is present in some complex systems, but advanced computer vision algorithms that are required for precise detection and dependable monitoring are often absent. Consequently, if they miss critical security events or generate an abnormally high number of false positives, their dependability and efficacy in real-world applications could be jeopardized.

2.2 Proposed System

The proposed system enhances security by providing consistent monitoring, even during the night, through the integration of real-time motion detection and night vision technology. It uses a night vision camera to continuously stream live video feeds. These streams are processed using OpenCV, a powerful computer vision toolbox, which accurately detects human motion while eliminating irrelevant background movements. This reduces the number of false alarms that go off and ensures that alerts only sound in response to genuine hazards. When it detects human presence, the device triggers a prerecorded audio alarm to notify nearby individuals or security personnel. This makes a quick response possible. This technology is flexible enough to accommodate various conditions while guaranteeing effective and continuous observation.

2.3 Paper Survey

1. S. Subha, Rahul, J. R. K, D. K, Yuvaraj and S. K. Mudradi, "Visual Object Detection in Extreme Dark Condition," *2023 3rd International Conference on Innovative Practices in Technology and Management (ICIPTM)*, Uttar Pradesh, India, 2023, pp. 1-5, doi: 10.1109/ICIPTM57143.2023.10117607. in thus paper to Examining the difficulties and developments in object detection in low light is the paragraph's primary goal. The text emphasizes the possible advantages of utilizing object detection techniques for night vision surveillance. These benefits include the ability to detect motion using PIR sensors and the enhancement of detection accuracy through the use of Deep Learning (DL) algorithms. The text also highlights how alert systems, such email alerts and GSM messages, can be integrated with video clips to improve security measures. The ultimate goal is to progress the field's understanding and refine techniques for object detection in low-light conditions.

2. Kamalakar Desai, R. R. (January 2024). Smart Night Vision camera system with Distress Identification and warning alert mechanism to notify crime. The primary goal is to outline the creation and characteristics of a cutting-edge night vision camera system intended to improve security and safety in places with a lot of nighttime activity. To enhance visibility and monitoring in low light, this system combines multiple technologies, such as long-wave infrared image fusion and shimmering night vision. In addition, it has features like self-cleaning lenses, crime zone alerts, real-time facial recognition, and audio sensors to identify distress signals and suspicious activity. The objective is to deliver a complete and effective security system that improves nighttime environment safety while addressing the shortcomings of conventional CCTV cameras.

3. Raza, A., Chelloug, S. A., Alatiyyah, M. H., Jalal, A., & Park, J. (2023). Multiple Pedestrian Detection and Tracking in Night Vision Surveillance Systems. *Computers, Materials & Continua/Computers, Materials & Continua (Print)*, 75(2), 3275–3289. <https://doi.org/10.32604/cmc.2023.029719>. The paper's primary goal is to introduce an innovative surveillance system made for tracking and detecting pedestrians in poor light and at night. The system makes use of image processing and machine learning methods, such as particle filters for tracking, random forest classifiers for picture segmentation, and infrared (IR) imaging. It exhibits great accuracy in segmentation, detection, and tracking and attempts to overcome the difficulties of identifying and tracking several pedestrians in difficult situations. The system's goal is to improve the efficiency of surveillance in infrared video surveillance applications, such as pedestrian monitoring, crowd analysis, and human detection.

4. Shelke, R. R. (2024). Active Security Surveillance and Object Detection. *International Journal of Information Technology and Computer Engineering*, 45, 38–47. <https://doi.org/10.55529/ijitc.45.38.47>. The paper's study to clarify how object detection systems and active security surveillance enhance real-time threat detection and monitoring. These systems track and identify suspicious objects or actions using cutting edge technology like deep learning, facial recognition, and motion detection. They improve response times and security, but they also bring up ethical and privacy issues that must be properly addressed. Sustained research is necessary to advance these technologies and solve associated problems.
5. Wnag, Y. K., & Fan, C. T. (2010). Moving Object Detection for Night Surveillance. <https://doi.org/10.1109/iihmsp.2010.66>. In low light, where typical background subtraction algorithms fail, the paper aims to present a novel approach to autonomous visual surveillance. The illumination-invariant background reduction and illumination compensation employed in the suggested method tackle problems including high noise levels, low contrast, and low brightness. According to experimental results, this method outperforms previous methods in identifying moving objects at night by a large margin.
6. Kunja Bihari Swain, D. Behera and S. S. Solanki, "Virtual instrumentation based night security system using speedy-33," *2014 IEEE International Conference on Advanced Communications, Control and Computing Technologies*, Ramanathapuram, India, 2014, pp. 749-753, doi: 10.1109/ICACCCT.2014.7019190. In this paper is to present an overview of a LabVIEW-developed security system that records and displays images, sounds, and video of a monitored area. The system consists of a basic robot with an obstacle sensor, a wireless transmitter/receiver for motor control, and a wireless infrared night vision camera. The robot's movements is controlled and the signals are processed using a DSP processor (NI Speedy 33). Using LabVIEW makes learning, hardware interface, and peripheral device access easier.
7. S. Parveen and J. Shah, "A Motion Detection System in Python and Opencv," *2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, Tirunelveli, India, 2021, pp. 1378-1382, doi: 10.1109/ICICV50876.2021.9388404. The primary goal of the study is to introduce a motion detection software system that monitors movement around an object or visual area in order to increase security. By employing Python-based software to detect and track items' movement and the amount of time they spend close to the camera, the system improves on already-existing security systems, such as motion sensor lighting and security cameras.

8. Chen, Y. L., Wu, B. F., Huang, H. Y., & Fan, C. J. (2011). A Real-Time Vision System for Nighttime Vehicle Detection and Traffic Surveillance. *IEEE Transactions on Industrial Electronics*, 58(5), 2030–2044. <https://doi.org/10.1109/tie.2010.2055771>. In this paper's is to build a traffic surveillance system that can identify and follow moving cars in traffic scenes at night. The system recognizes headlights and taillights on vehicles using pattern analysis and image segmentation. To manage different evening lighting situations, it uses an automatic multilevel histogram thresholding in conjunction with a rapid bright-object segmentation technique. After that, the system analyzes vehicle light patterns and classes vehicles using spatial grouping and tracking. The system has been successfully tested for real-time vehicle recognition and identification in various nighttime conditions on an embedded platform based on the TI DM642 DSP.

9. Ahire Upasana, B. M. November 2015. Real Time Security System using. *International Journal of Computer Science and Mobile Computing*. in this paper to develop a security system that alone captures and notifies when it detects motion. Using a webcam, this system will keep an eye on places that are off-limits, capture images only in response to movement, sound an alarm, and notify the manager and police by SMS. This method offers fast security hazard alerts while also conserving storage.

10. Shahid, M. F., Shah, A., Bashir, M. W., & Ikram, M. S. (2023). Low-Cost Multi-Layer Approach for Motion Detection System for Area Monitoring. <https://doi.org/10.1109/icosst60641.2023.10414204>. in this paper to improve the Doppler-based movement detection of the RCWL-0516 radar module by giving it directionality and the ability to provide velocity and other information using the FFT technique. For better motion detection accuracy, the system combines this upgraded radar with a camera and a passive infrared (PIR) sensor. It also has anti-jamming features, GSM real-time notifications, and audio-visual alarms. The integrated strategy seeks to enhance safety, automation, and security across a range of applications.

SYSTEM REQUIREMENTS

3.1 Hardware Requirements

- RAM : 8 GB or more
- Camera : Night Vision Camera
- HDD : 40 GB more
- Processor : Intel I Core or higher

3.2 SOFTWARE REQUIREMENTS

- Operating System : Windows 10 or Linux
- Versions of Python : 3.76.X or latest
- Supporting software : Anaconda
- IDE : Visual Studio

3.3 Technologies Overview

3.3.1 Visual Studio

An integrated development environment (IDE) is a feature-rich program that helps with many aspects of software development. As a creative launchpad, the Visual Studio IDE allows you to modify, debug, and create code as well as publish an application. In addition to the typical editor and debugger included in most integrated development environments (IDEs), Visual Studio also includes compilers, code completion tools, graphical designers, and a plethora of other features to enhance software development.

Key Features of the Visual Studio is as follows:

- **Coding Languages:** Visual Studio offers an integrated environment for writing, editing, and debugging several programming languages such as C++, C#, JavaScript, TypeScript, Python, and more. The process of creating new features and fixing faults is streamlined by the inbuilt debugger.
- **Extensive IDE:** A wide range of tools are available in Visual Studio, beyond the conventional debugger and editor. These include things like compilers, code completion tools, graphical designers, and other enhancements that make software development more productive.
- **Cross-Platform Development:** Visual Studio allows you to create apps for multiple platforms. Regardless of whether you're targeting cloud, desktop, online, or mobile services, the IDE provides you with the necessary tools and frameworks.
- **Version Control Integration:** Integrate version control tools, like Git, to facilitate easy team collaboration. Continue branching, monitor changes, and work well with others.
- **Installer Based on Workload:** To customize your installation, simply select the components that are required for the specific development activities you are working on. This ensures a straightforward configuration that satisfies your needs.

3.3.2 HTML

Web pages are created using a language called HTML, or HyperText Markup Language. It uses HTML tags and properties to define a web page's structure and layout. Search engines follow the components that make up HTML to determine how to display material on websites. Examples include headings, lists, images, links, and more.

Features of HTML:

- It's easy to learn and use.
- Platform independence is a feature.
- You can add images, videos, and audio to a webpage.
- Hypertext may be added to the text.
- It's a markup language.

Structure of an HTML Page

- The document type declaration, or `<!DOCTYPE html>`, is technically not a tag. It shows that the document in question is an HTML document. The doctype declaration does not include case sensitivity.
- `<html>` The HTML root element is the name given to this element. It includes every other element.
- `<head>` - The head tag contains the "behind the scenes" information for a webpage. Elements inside the head are not shown on the front end of a webpage. The `<head>` element contains the following HTML elements:
 - `<style>` - We may add styling and aesthetic appeal to our webpages by using this HTML tag in conjunction with CSS.
 - The title of the page you are currently seeing appears at the top of your browser when you visit a website.
 - For each relative URL in a document, the root URL is indicated via the `<base>` element.
 - `<script>`: This webpage tag has additional functionality thanks to JavaScript.
 - `<meta>`: Every time a page is browsed, the metadata for the website has to load from this tag. For instance, you may use the metadata charset to encrypt your website using the normal UTF-8 encoding. As a result, users can view your webpage in their favorite language. The tag closes on its own.
 - `<link>` - The 'link' tag connects JavaScript, HTML, and CSS. It closes by itself.
 - `<body>`: The body tag contains the visible content of a webpage. Put another way, the body content is the information that shows up on the user's browser.

3.3.3 CSS

Web page presentation was made simpler with the creation of the CSS (Cascading Style Sheets) language. It provides guidelines on the colors, fonts, spacing, and placement that should be utilized to characterize a webpage's design and allows you to apply styles to HTML documents. Thanks to CSS, developers and designers have a great deal of control over how HTML components are displayed.

HTML employs tags, but CSS makes use of rule sets. To add CSS styles to the HTML element, selectors are utilized. While CSS is easy to learn and understand, it provides a great deal of flexibility over how an HTML document is displayed.

Why is CSS used?

- saves time by just needing to create CSS once and then applying it to several HTML pages.
- Easy Maintenance: A single alteration can alter the style worldwide.
- Methods of clean coding that improve readability for search engines are called "search engine friendly."
- Improved Styles: Compared to HTML, more properties are available.
- Offline Browsing: Web applications can be stored locally by CSS using offline caching, allowing for offline viewing.

Types of CSS:

1. Inline CSS: Inline CSS directly adds styles to particular HTML components by using the style tag. This approach allows unique styling of HTML document components, overriding external or internal styles.
2. Internal or Integrated CSS: The definition of internal or embedded CSS is contained in the HTML document's <style> element. It does this by applying styles to particular HTML elements. The <style> tag contains the CSS rule set, which is located in the head section of the HTML file.
3. External CSS: External CSS is made up of separate CSS files that only include style properties (class, id, heading, etc.) and is applied via tag attributes. The CSS property is written in a separate file with the .css suffix and needs to be linked to the HTML content using a link tag. It means that all web pages will use the same style, which can only be set once for each element.

Advantages of CSS

- You can save time by creating CSS only once and using it on multiple HTML pages. Every HTML element has a style that you can specify and apply to as many Web pages as you'd like.
- Faster page loads: You can avoid constantly adding HTML tag characteristics by using CSS. To apply a CSS rule to every instance of a tag, all you need to do is create one for each tag. Faster download times are therefore correlated with less code.
- Simple maintenance: Any changes made to a style will instantly update all elements across all web pages. This facilitates the process of making global modifications.

3.3.4 Python

Python is a high-level, interpreted, general-purpose dynamic programming language that prioritizes legible code. Its programs are typically smaller than those written in C and Java. It was founded in 1991 by developer Guido Van Rossum. Python is one of the most popular and rapidly growing languages in the world. Python is an effective, versatile, and easy-to-use language. The Python community is likewise thriving. Many businesses use it because it supports multiple programming paradigms. It also automatically maintains memory.

What possibilities can Python provide?

- Python can be used on a server to create web applications.
- Python can be used in conjunction with other technologies to develop workflows.
- Python can communicate with database systems. It is capable of reading and editing files.
- Python makes it feasible to do complex mathematical operations and handle large amounts of data.
- Python can be used for rapid prototyping and for creating software that is ready for production.

For what purpose, Python?

- Python is supported on a wide range of platforms, including Windows, Mac, Linux, Raspberry Pi, etc.
- The syntax of Python is simple and similar to that of English.
- Compared to certain other languages, Python has a syntax that allows programmers to write programs in less lines.
- Python can be worked with in a procedural, object-oriented, or functional manner.

Characteristics of Python

1. Free and open source: You may get the Python language for free and open source from the official website. All you have to do to obtain it is click the keyword "Download Python" and utilize the given download link. Get Python from this link. The source code is also accessible to the general public because it is open-source. You can download, use, and distribute it as a result.
2. Simple to write: Python is one of the higher-level programming languages. Python is quite easy to learn compared to other programming languages like C, C#, Javascript, Java, etc.

Python is a somewhat easy language to code with; anyone can learn the basics in a few hours or days. The language is also developer-friendly.

3. Easy to Read: You will find that learning Python is quite simple. As was already mentioned, Python has a fairly simple syntax. The indentations define the code block instead of semicolons or brackets.
4. Using Language and Objects: One of the main features of Python is object-oriented programming. Classes, object encapsulation, object-oriented languages, and related concepts are all supported by Python.
5. GUI Programming Support: Graphical user interfaces can be made with Python by using modules such as PyQt5, PyQt4, wxPython, or Tk. PyQt5 is the most popular tool used by Python programmers to construct graphical applications.
6. Proficient Linguistics: Python is one high-level language. System design and memory management are not necessary to learn when writing Python programs.
7. Strong Community Campaigning: Python has grown in popularity over time. Our queries are always met with insightful answers from the large StackOverflow community. When looking for solutions to frequently asked issues about Python, Python users can visit these websites.
8. Easily Fixed: Excellent information for locating mistakes. Most of them should be easy for you to identify and correct.

The advantages of Python as a programming language are as follows:

- Third-party module ecosystem: Python has a strong third-party module and library ecosystem that enhances its functionality for a variety of applications.
- Huge support library: Python's extensive support library, which include Pandas for data analytics and NumPy for numerical computations, makes it suitable for scientific and data-related applications.
- Being open source and with a large, active community: Python is both open source and has a large, active community that helps to further its development.
- Versatile, easy to learn, write, and read: Python's these attributes make it a popular programming language suitable for both beginners and specialists.

3.4 Software Requirements Specification

software requirements A specification is a written document that enumerates all of the features and capabilities of a software program. It is sometimes called a system requirements specification. It is broken up into a number of sections and pieces that describe the performance and functionality that the system should have in order to satisfy user needs. In addition to specifying how the system should behave, the specification gives a high-level rundown of the main business processes that will be facilitated, the simplifying assumptions that have been applied, and the fundamental performance standards that the system needs to meet. This document is necessary to ensure that everyone participating in the development process is informed.

Why SRS needed?

1. Explicitly and Clearly Defined Requirements: An SRS provides a detailed description of the requirements for the behavior, features, and constraints that an application or system must follow. This clarity ensures that the development satisfies the user's or customer's needs.
2. Reference Point During Development: Throughout the system or application development process, the SRS serves as a point of reference to ensure consistency and adherence to the initial objectives by verifying that the requirements are being met.
3. Promoting Communication and Collaboration: By providing team members with a common vocabulary and terminology, the SRS facilitates more effective communication and collaboration. By doing this, miscommunications and disputes over the requirements of the system are reduced.

3.5 Functional Requirements

- Night Vision Capability: To record and broadcast live video even in total darkness, the system ought to have a night vision camera.
- Real-Time Motion Detection: By removing unnecessary background activity, the system must use the OpenCV library to detect human movement in real-time.
- Automated Alerts: The system should automatically play a pre-recorded audio file to sound an alarm when it detects human presence.
- Alert Management: When necessary, authorized users should be able to handle, accept, or reject notifications.

3.6 Methodology Adopted

1. **System Integration and Hardware Configuration:** The first step in the project is to integrate a night vision camera that can record sharp video in low light. A processing unit that facilitates real-time video analysis is connected to the camera.
2. **Processing of Live Video Feeds:** The processing unit continuously receives live video feeds that are continuously recorded by the night vision camera. These video frames are preprocessed using OpenCV, a potent computer vision toolkit, to improve image quality and eliminate noise that might compromise the precision of motion recognition.
3. **Motion Detection Implementation:** To detect human presence in real-time, sophisticated motion detection algorithms are created with OpenCV. To reduce the amount of background activity that is irrelevant to human movements, such as shadows, animals, or weather impacts, techniques including backdrop removal, frame differencing, and contour detection are used.
4. **Automated Alarm System:** When an automated alert system detects human movement, it immediately sounds a recorded audio alarm. This guarantees that everyone in the vicinity or security staff are immediately alerted to any possible security breach, enabling a timely response.
5. **Deployment and Optimization:** The system is deployed in the intended environments after testing is deemed successful. Ongoing tuning is carried out to enhance detection capabilities, minimize false alarms, and guarantee efficient round-the-clock monitoring.

3.7 Non-Functional Requirements

Non-functional requirements (NFRs) are specifications used in software engineering that enumerate the attributes and functionalities of a software system. They focus more on the way the system functions than on the specific tasks it accomplishes. NFRs are necessary to ensure the software system's overall usability, effectiveness, dependability, and maintainability.

Categories of non-functional requirements are:

- **Performance and scalability:** Required response times; benchmark specifications; how the system behaves under heavier loads.
- **Operating limitations** include software requirements, system requirements, and run-time limits.

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- Platform constraints: Clearly state any limitations or requirements particular to a certain platform.
- Modifiability: The effort needed to make changes to the software; helps with future change planning.
- Portability: The program may not be compatible with all hardware and operating systems, making cross-platform transfers difficult.
- Reliability: How frequently software faults occur, their impact, how to find them, and how to fix them.
- Security: Conditions for protecting the information and system; the gravity and length of a system breach; and methods for mitigating it.
- Usability: Improving the user's experience and making the system easier to comprehend and operate.

3.8 System Architecture and Design

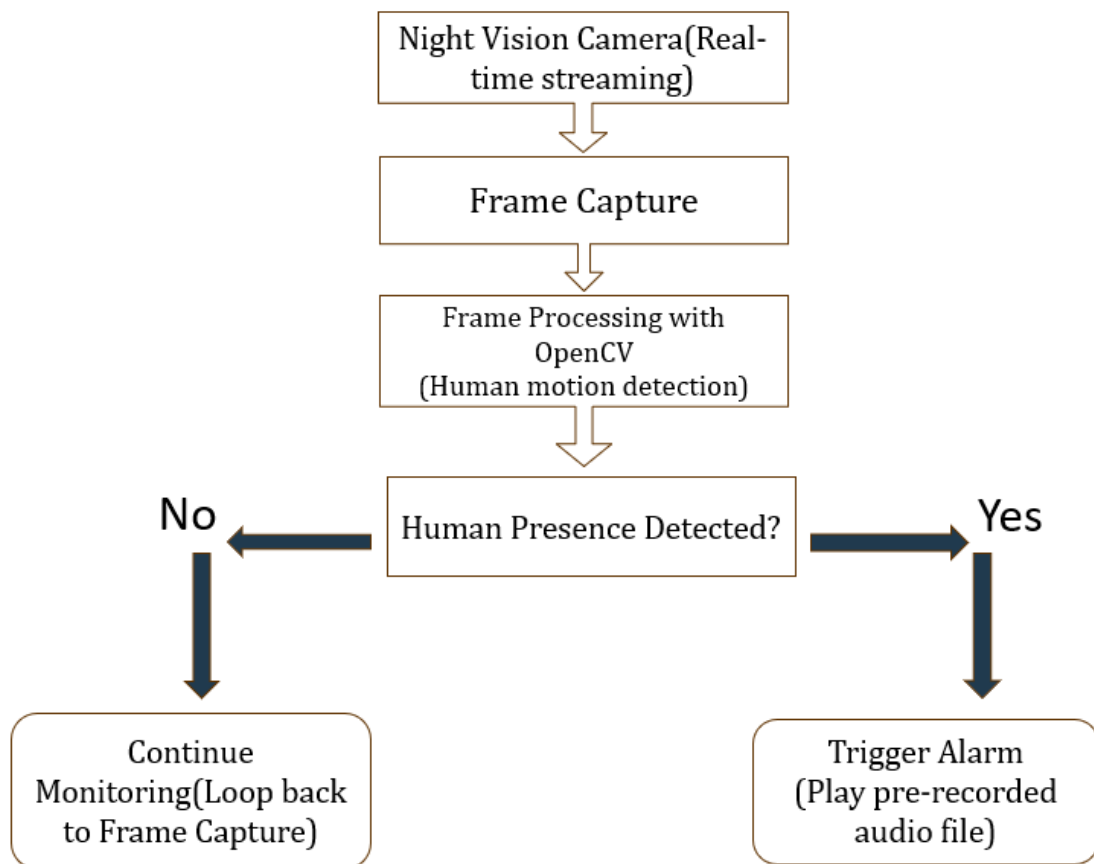


Figure 3.8: architecture diagram

- **Night Vision Camera:** Provides a live stream of video and records even in dimly lit environments. This part is essential to making sure the security system functions properly at night or in the dark surroundings.
- **Video Stream:** Provides processing with real-time video data from the camera. unit. acts as the image processing algorithms' input, enabling ongoing observation of the region.
- **OpenCV Frame Processing:** Makes use of OpenCV, an open-source to process individual video frames, using a computer vision library. applies a picture methods for processing to improve frame quality and get ready for motion analysis of detection.
- **Motion Detection Algorithm:** detects motion by analyzing processed frames. by contrasting successive frames and determining modifications that indicate motion. distinguishes between background motions and noteworthy motion that can be a sign of a person.

- **Trigger Alarm:** In the event that human motion is verified, this sets off a pre-recorded audio alarm. Notifies homeowners or security workers right away of any possible incursions.

3.9 Flow Chart

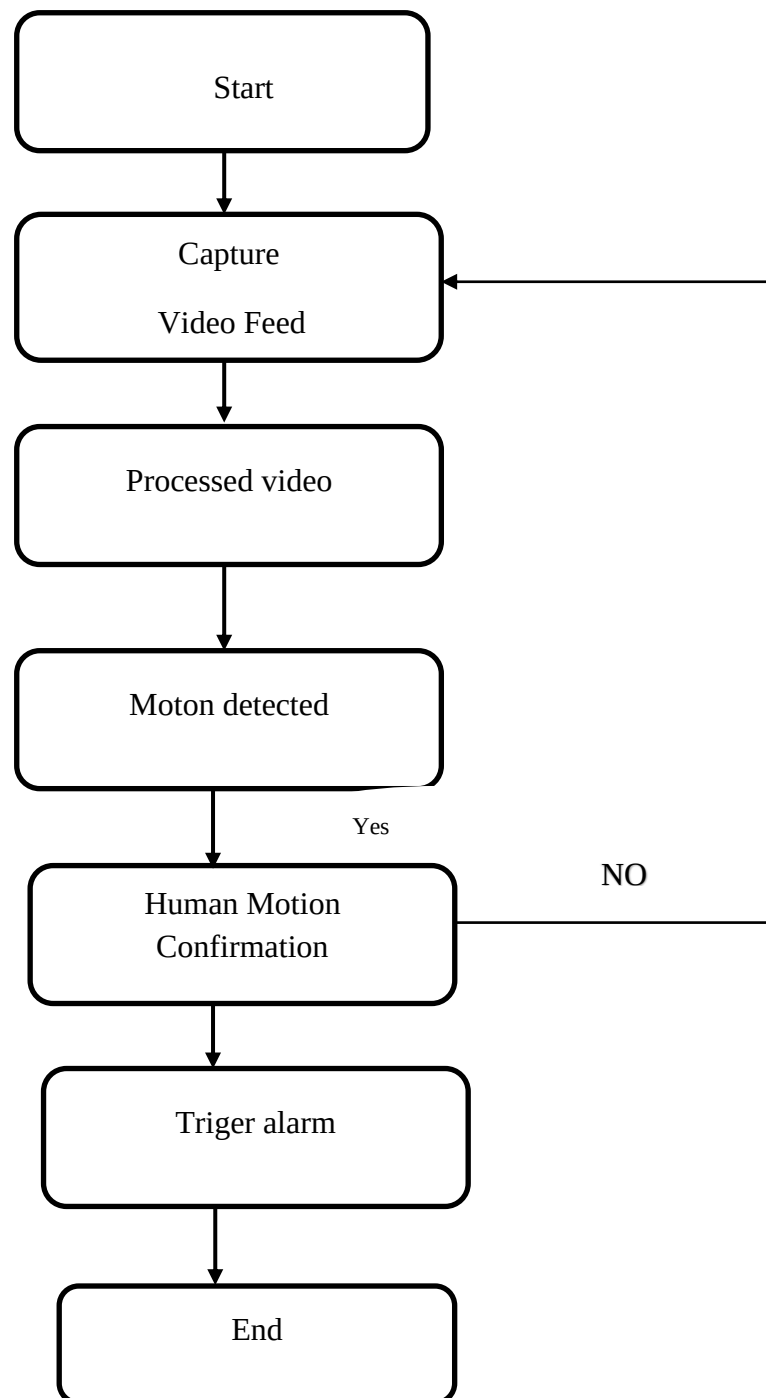


Figure 3.9: Flowchart of Night Vision Motion Detection

The monitoring system initially records and preprocesses the video signal from a night vision camera. Next, it looks for any movement in the video. If it detects no motion, it returns to

record additional footage. The system decides whether motion is detected and if it is human-generated. If not, it continues to record video. If the equipment detects that the motion is in fact human, it raises an alarm

Code:

```
import cv2

import imutils

import winsound

import tkinter as tk

from tkinter import messagebox

def start_detection():

    cap = cv2.VideoCapture(0)

    if not cap.isOpened():

        messagebox.showerror("Camera Error", "Could not open camera. Please check your camera settings.")

        return

    while True:

        ret, frame = cap.read()

        if not ret:

            messagebox.showerror("Camera Error", "Failed to capture video.")

            break

        frame = imutils.resize(frame, width=650)

        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

        face_cascade = cv2.CascadeClassifier(cv2.data.harcascades + 'haarcascade_frontalface_default.xml')

        faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30, 30), flags=cv2.CASCADE_SCALE_IMAGE)

        face_count = 0

        for (x, y, w, h) in faces:
```

```
cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)

face_count += 1

if face_count > 0:

    winsound.Beep(1000, 200)

    cv2.putText(frame, "Total humans detected = " + str(face_count), (10, 425),
cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)

    cv2.imshow("Frame", frame)

    key = cv2.waitKey(1)

    if key != -1:

        break

cv2.destroyAllWindows()

cap.release()

login_window.deiconify()

def login():

    username = entry_username.get()

    password = entry_password.get()

    if username == "admin" and password == "pass":

        show_success_window(username)

    else:

        messagebox.showerror("Login Failed", "Invalid username or password.")

def show_success_window(username):

    success_window = tk.Toplevel()

    success_window.title("Login Successful")

    success_window.geometry("400x400")

    success_window.configure(bg='#90EE90')

    success_label = tk.Label(success_window, text=f"Welcome, {username}!", font=("Arial",
30), bg='#90EE90')

    success_label.pack(expand=True)
```

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```
name_label = tk.Label(success_window, text="Name: Supriya Badami", font=("Arial",
20), bg='#90EE90')

name_label.pack(pady=(20, 5))

reg_num_label = tk.Label(success_window, text="Reg No: [P15ZZ22S126017]",
font=("Arial", 20), bg='#90EE90')

reg_num_label.pack(pady=(0, 40))

ok_button = tk.Button(success_window, text="OK", font=("Arial", 20),
command=lambda: [success_window.destroy(), login_window.withdraw(), start_detection()])

ok_button.pack(pady=20)

def exit_application():

    login_window.destroy()

login_window = tk.Tk()

login_window.title("Sign-In Page")

login_window.geometry("600x500")

login_window.configure(bg='#ADD8E6')

login_window.grid_columnconfigure(0, weight=1)

login_window.grid_columnconfigure(1, weight=1)

university_heading = tk.Label(login_window, text="RANI CHANNAMMA UNIVERSITY",
font=("Algerian", 45), bg='#ADD8E6')

university_heading.grid(row=0, column=0, columnspan=2, padx=20, pady=10,
sticky="nsew")

project_heading = tk.Label(login_window, text="Welcome to My Project", font=("Algerian",
35), bg='#ADD8E6')

project_heading.grid(row=1, column=0, columnspan=2, padx=20, pady=10, sticky="nsew")

tk.Label(login_window, text="Username:", font=("Algerian", 30),
bg='#ADD8E6').grid(row=2, column=0, padx=10, pady=(10, 5), sticky="e")

entry_username = tk.Entry(login_window, font=("Arial", 30))

entry_username.grid(row=2, column=1, padx=20, pady=5)

tk.Label(login_window, text="Password:", font=("Algerian", 30),
bg='#ADD8E6').grid(row=3, column=0, padx=10, pady=(10, 5), sticky="e")
```

```
entry_password = tk.Entry(login_window, show="*", font=("Arial", 30))

entry_password.grid(row=3, column=1, padx=20, pady=5)

login_button = tk.Button(login_window, text="Login", font=("Arial", 30), bg='#90EE90',
command=login)

login_button.grid(row=4, column=0, columnspan=2, padx=20, pady=20, sticky="nsew")

exit_button = tk.Button(login_window, text="Exit", font=("Arial", 30), bg='#FF6347',
command=exit_application)

exit_button.grid(row=5, column=0, columnspan=2, padx=20, pady=20, sticky="nsew")

login_window.mainloop()
```

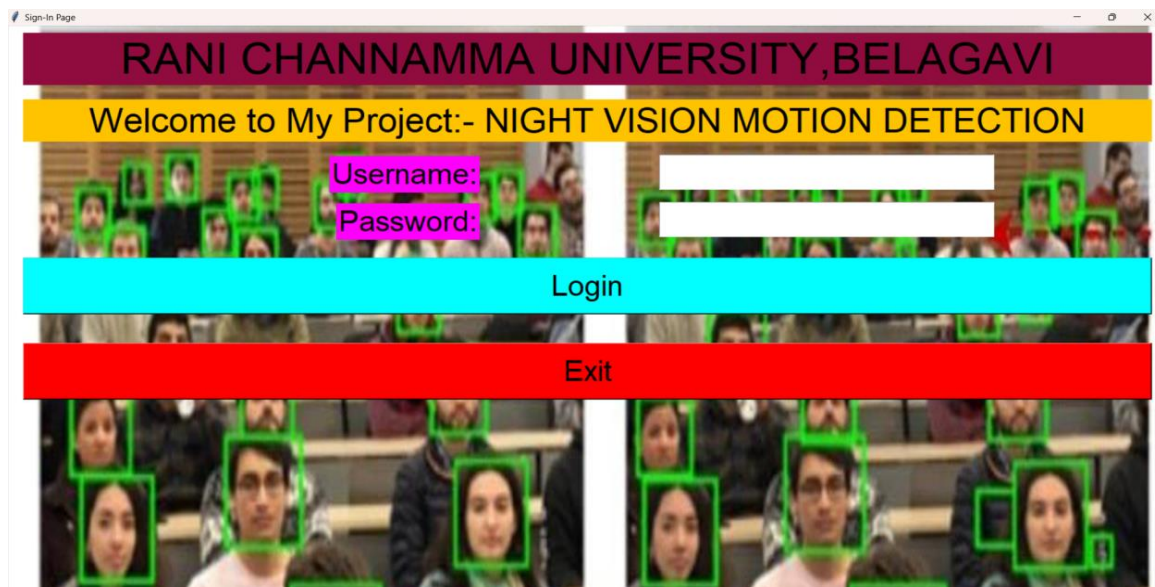
Methodology used in code:

- Face Detection with OpenCV: Uses the Haar Cascade classifier (haarcascade_frontalface_default.xml) to detect faces in real-time from a video stream. Frames are converted to grayscale for faster processing, and rectangles are drawn around detected faces.
- Tkinter for GUI: Provides a graphical user interface for login, with fields for username and password. On successful login, a custom "Login Successful" window displays a personalized message with the user's name and registration number.
- Real-Time Video Processing: Captures video frames from the webcam, resizes them for efficiency, and processes them frame-by-frame to detect faces continuously.
- Sound Alerts (winsound): Plays a beep sound when a face is detected, providing real-time audio feedback.
- Login Logic: Simple username and password verification, with feedback through success and error messages. A new window opens on successful login, which triggers face detection.

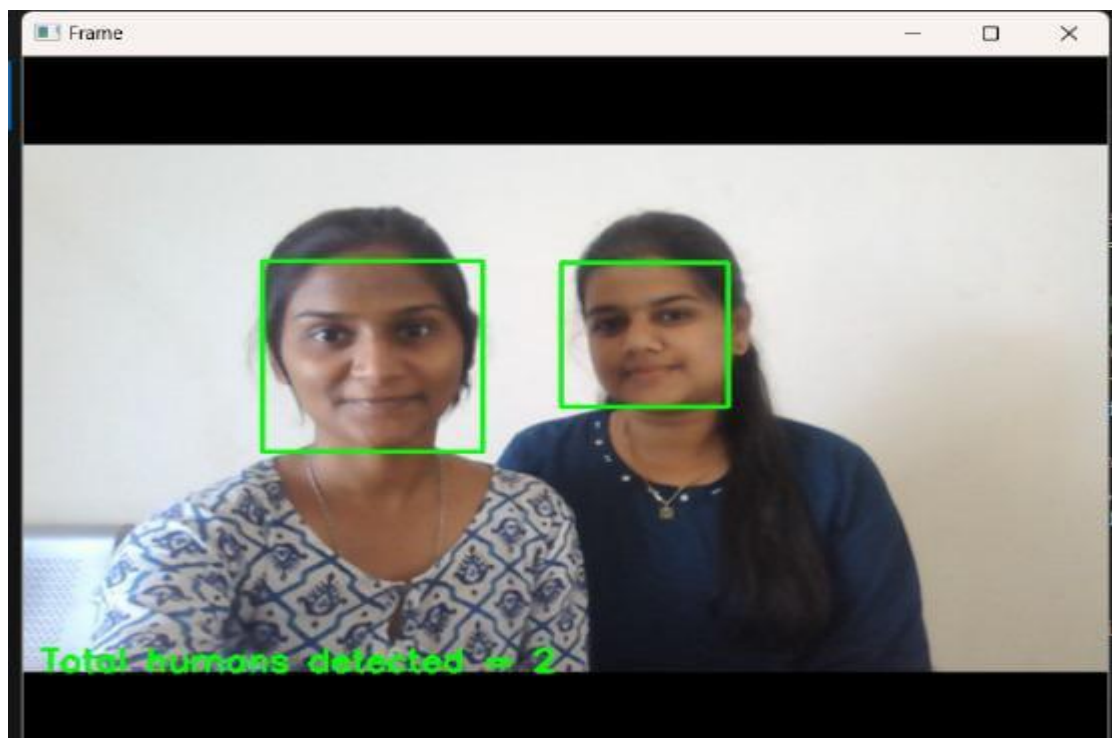
IMPLEMENTATION AND RESULTS

4.1 Implementation Output Screenshots

a) Result 1:

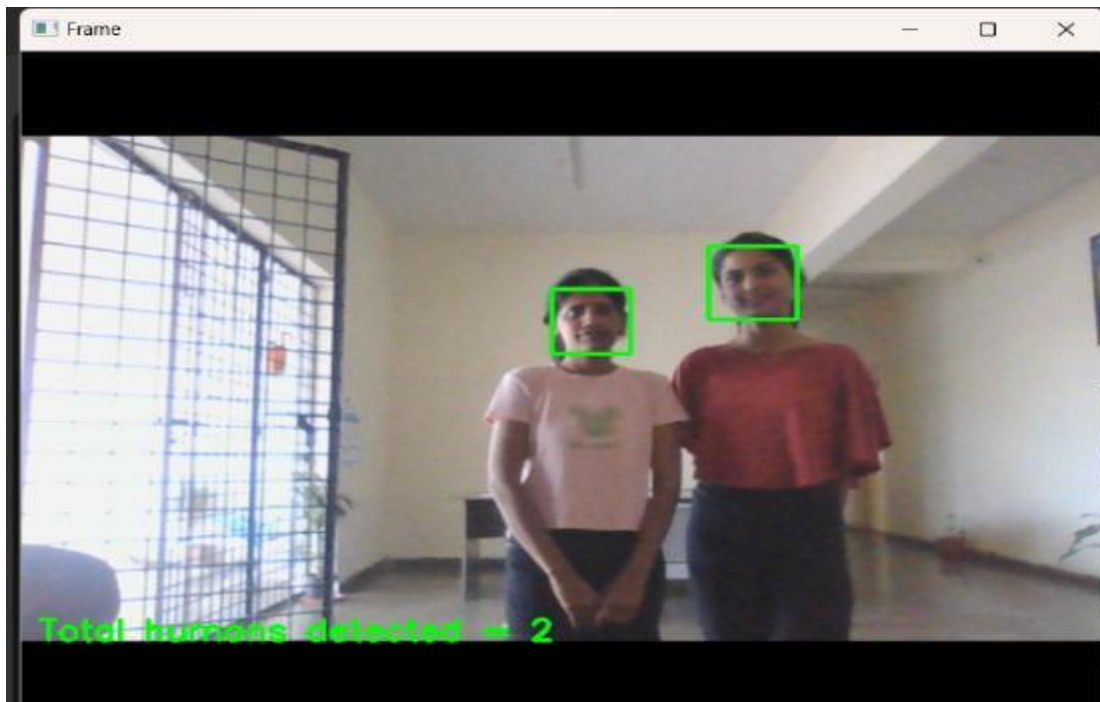


Sign in Page



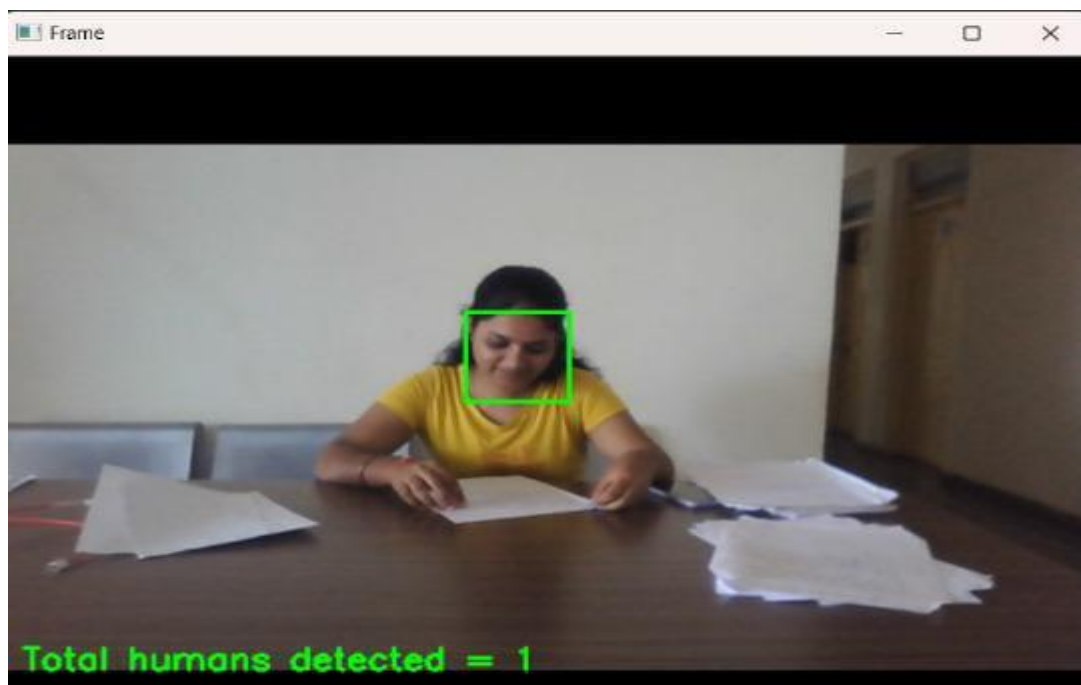
Total human detected=2

b) Result 2:



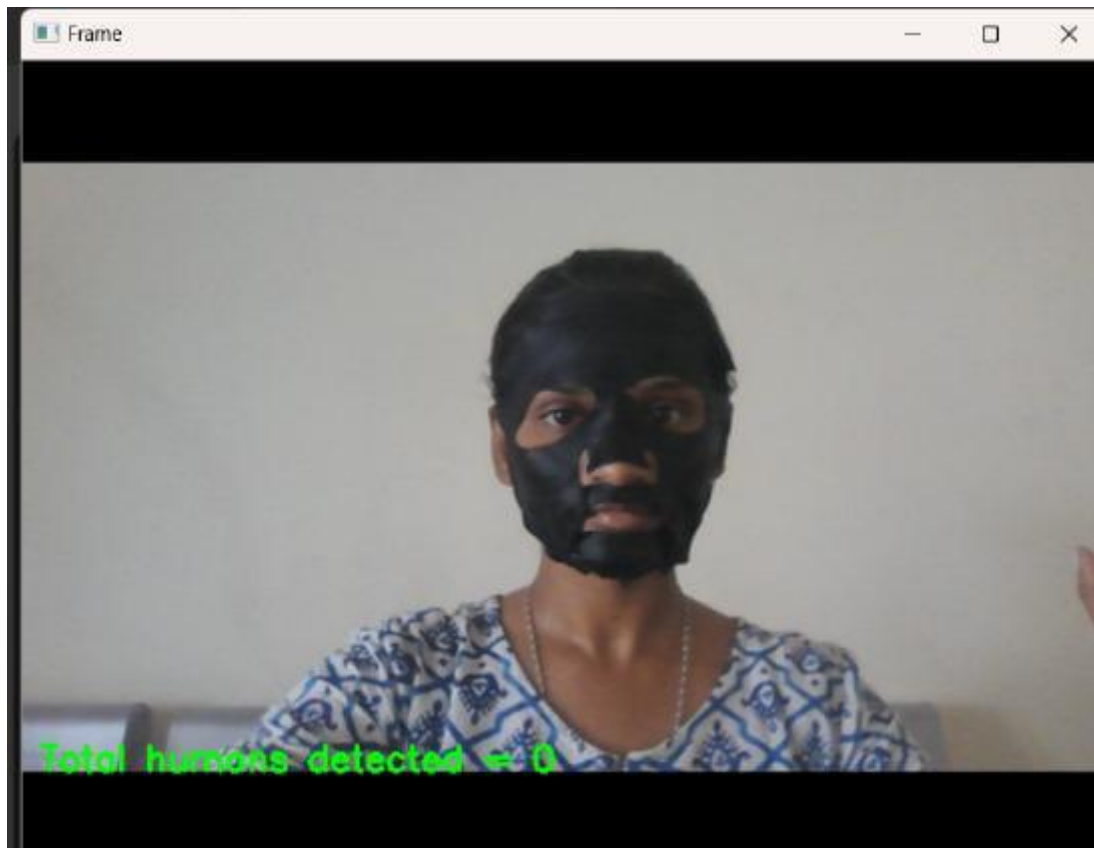
Total human detected=2

b) Result 3:



Focus of the camera detected

b) Result 4:



Total human detected=0

Reason:- Total human detected is 0 because the face is covered.

b) Result 5:



In this result you can see the distance of the camera is placed to capture the motion.

CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

The "Night Vision Motion Detection" project aims to transform surveillance systems, particularly in poorly illuminated environments. The system ensures reliable, constant surveillance by using live video feed from state-of-the-art night vision cameras, regardless of the lighting. Integrating frame processing with OpenCV allows for precise identification of human motion while eliminating distracting ambient noise, significantly reducing the quantity of false alarms. Additionally, the automated alarm system starts an audio alarm that has been pre-recorded to ring when it detects human presence and instantly notify security workers who could be dangerous. This program provides a cutting-edge solution for modern security barriers, including effective motion detection in low-light conditions. It enhances safety and security in a range of settings, such as the home, workplace, and public areas. This project's technological innovations promise to provide better security and peace of mind while setting a new benchmark for surveillance systems.

5.2 Future Enhancement

1. Classification of AI Objects: Include cutting-edge AI to recognize and distinguish between objects (humans, animals, etc.).
2. Cryogenic Imaging: To improve detection in total darkness, combine infrared cameras with night vision.
3. Tailored Alerts: Permit consumers to customize their alert options and get email or SMS notifications.
4. At-A Distance: Create applications for system control and monitoring from a distance.
5. Scalability: Create a system that can grow and is appropriate for bigger spaces.

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APPENDIX

Referred Books

- The complete reference python: By Martin C. Brown.
- Python machine learning: By Sebastian Raschka & Vahid Mirjalili.

Referred website

- https://www.researchgate.net/publication/366333281_Subscribers_Forecasting_of_Netflix_Based_on_Multiple_Linear_Models
- <https://www.sciencedirect.com/science/article/abs/pii/S0040162522007399>
- <https://www.programcreek.com/python/example/89428/cv2.absdiff>
- <https://nypost.com/2024/04/18/media/netflix-crushes-subscriber-forecasts-again-with-more-than-9m-new-customers/>